

From Physiological Allostasis to Artificial Risk Regulation: A Multi-Timescale Biomimetic Proof of Concept Using Three Decades of XCMG Prices, Drawdowns, and Recovery Dynamics

Biomimetic Allostatic Regulation in Nonstationary Markets

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Abstract

Financial forecasting systems commonly prioritize directional classification or point-forecast accuracy, yet capital processes operating in nonstationary markets must also withstand breaches of risk boundaries and recover from recurrent shocks. Physiological allostasis offers a different design logic: anticipatory regulation, coordination across fast and slow timescales, accumulation of regulatory load, and path-dependent recovery. This study translates these functions into a Biomimetic Allostatic Risk and Viability Controller (BARVC). It conducts a longitudinal, single-security proof of concept using 7,109 daily observations for XCMG Machinery from August 28, 1996, through August 12, 2026. The sample contains 91 distinct drawdown cycles, including a historical maximum drawdown of 89.66%, and 124 major upward-move events. Model development used 1996–2018 for training, 2019–2021 for calibration, and locked 2022–2026 data for evaluation, with a 20-trading-day purge window separating calibration from testing. In the locked test period, the full-state model achieved 72.02% coverage for the P10–P90 interval, with an absolute coverage error of 7.98 percentage points; the corresponding non-allostatic model achieved 75.35% coverage and a 4.65-point error. Relative to buy-and-hold, BARVC reduced maximum drawdown from 36.08% to 21.77% and the daily severe-breach rate from 16.16% to 0.83%. However, a constant 45% risk exposure produced a lower maximum drawdown of 17.41% and no severe breaches. The empirical results supported none of the five prespecified mechanistic advantage hypotheses. The study therefore establishes the computational feasibility and falsifiability of allostatic functional translation but does not establish superior investment performance. Cross-security validation and a larger sample of independent stress events are required to determine whether the observed findings generalize beyond the focal security.

Keywords: biomimetics; allostasis; allostatic load; anticipatory regulation; multi-timescale control; recovery memory; quantile boundaries; nonstationary systems

1. Introduction

1.1. The Problem of Risk Regulation in Nonstationary Systems

Financial markets are not simply stable processes overlaid with random noise. Conditional means, conditional variances, tail distributions, liquidity, and the relationships between predictors and outcomes may all change over time. Volatility clusters rather than remaining constant [1,2]; latent regime shifts can alter the dynamics of returns and risk simultaneously [3,4]; and structural breaks can invalidate parameters that appeared stable in earlier samples [5,6]. Extreme shocks and prolonged underwater periods add a path dimension that variance alone cannot express. Even after a tolerable one-day loss, a capital process may remain below its prior peak for years, weakening risk-bearing capacity, compound growth, and resistance to subsequent shocks. Research on resilience likewise emphasizes that preserving function and recovering capacity are distinct from static stability [7,8].

This distinction changes the engineering character of the problem. Market prices are external environmental variables that a portfolio controller cannot directly govern. What an artificial system can regulate is risk exposure, the risk budget, defensive state, boundary width, and the speed of re-entry into risk. Price or return forecasts may inform these actions, but forecasting is not regulation. Point forecasts, in particular, compress the range of possible future outcomes and may be least informative precisely when state uncertainty is greatest. For a system exposed to nonstationary shocks, the more defensible objective is to maintain viability: limit severe breaches of the risk boundary, preserve deployable capital, and restore functional capacity after stress.

The issue is not that financial markets are never predictable. Predictability is regime-dependent, horizon-dependent, and model-dependent. Nonlinear methods can extract economically meaningful patterns from large sets of predictors [9], and volatility is generally more predictable than returns. Forecast combinations may improve robustness in some environments [10], but the risk of breaking still weakens model transportability [11]. Superior in-sample fit does not automatically create out-of-sample value: predictive relationships can decay, model selection can introduce optimism, and small advantages can be erased by distribution shift or execution costs [12-14]. The more consequential design question is therefore whether a control architecture can incorporate predictive information while preserving interpretability, boundedness, and falsifiability in the event of forecast failure.

1.2. Limitations of Conventional Forecasting and Static Risk Control

This study is motivated by five interrelated limitations. First, in-sample performance may differ substantially from genuine forward performance. Overfitting, repeated specification search, parameter instability, and data leakage can cause a model to learn historical accidents that do not persist [12-15]. The appropriate response is temporal isolation, purging of labels that overlap period boundaries, freezing of model decisions, and simultaneous use of strong and weak benchmarks—not still more elaborate retrospective fitting.

Second, daily point forecasts do not fully represent decision uncertainty. They compress a potentially asymmetric or even multimodal future distribution into a single value, and their evaluation depends on the target functional and the loss function [16]. Conditional quantiles, conditional risk models, and strictly proper scoring rules provide decision-relevant descriptions [17-22]. Still, nominal coverage alone remains insufficient: an excessively wide interval can be well calibrated while offering little control value. A valid evaluation must therefore consider both calibration and interval sharpness.

Third, static risk thresholds may be mismatched to a changing environment. A fixed volatility target, drawdown trigger, or risk budget may react too slowly after a rapid shock, remain excessively defensive after stabilization, or restore exposure before latent vulnerability has dissipated. Volatility-managed portfolios demonstrate the potential

value of state-dependent exposure [23], but volatility is only one dimension of stress. Drawdown depth, stress duration, recurrent breaches, and failed recovery contain additional path information.

Fourth, conventional drawdown statistics are important but incomplete. Maximum drawdown summarizes the most severe peak-to-trough loss, while conditional drawdown measures extend attention to the tail of the drawdown distribution [24-27]. These measures do not themselves define a regulatory state that accumulates repeated stress, remembers failed recovery, and constrains later risk re-entry. A controller requires both a risk-measurement layer and a state-transition layer.

Fifth, biological terminology alone does not make an algorithm biomimetic. A defensible mechanism translation requires, at minimum, an identifiable biological source, physiological function, abstract design principle, artificial state variable, computational module, control consequence, and falsification experiment [28-34]. Without this chain, biological language adds narrative complexity rather than testable content. Mechanism deletion, mechanism reversal, and complexity-matched nonbiological controls are therefore indispensable.

These limitations do not require abandoning forecasting. They require repositioning it within a broader systems architecture that accumulates stress, changes control objectives across timescales, and constrains the recovery path.

1.3. Theoretical Value of Physiological Allostasis

Homeostasis and allostasis should not be treated as mutually exclusive theories. Homeostasis describes the maintenance of regulated variables within functional bounds through feedback processes [35,36]; it does not imply absolute stasis. Allostasis further emphasizes that a system can alter regulatory activity, reference levels, and resource allocation in response to anticipated demand, current state, and prior stress to preserve higher-order function [36-39]. The two can be hierarchically nested: homeostatic feedback operates locally, while allostatic coordination changes when and how strongly that feedback is invoked.

Five functions are particularly relevant to artificial risk systems. First, stability is maintained through change. Biological systems do not hold every variable at a single value; they reallocate resources to sustain organism-level function. In an artificial capital system, the object of regulation is not the market itself but risk exposure and risk capacity.

Second, allostasis is anticipatory. Organisms use contextual cues and memory to prepare resources before demand is fully expressed [38-41]. The corresponding artificial mechanism uses a conditional risk distribution to contract the risk budget before realized losses cross a feedback threshold. This does not imply precise price prediction. It means that uncertainty forecasts may have regulatory value when their role is constrained and empirically testable.

Third, physiological regulation operates across multiple timescales. Fast responses address acute disturbances, whereas slower processes alter baselines, sensitivity, and resource allocation. This study maps these functions to a fast market-stress loop and a slow structural-state loop. The selected financial windows are testable computational assumptions, not constants derived from human physiology.

Fourth, recurrent or inefficient regulation produces allostatic load: an accumulated cost that changes how the system responds to the next shock [42-44]. In the artificial system, load is constructed from volatility stress, drawdown depth, time underwater, quantile breaches, and failed recovery. These financial variables are not direct substitutes for cortisol, heart rate, neural activity, or other physiological measures. They are artificial system state variables that structurally correspond to biological regulatory functions. The mapping is functional, not homologous.

Fifth, recovery is path-dependent. Removal of an external stressor does not mean that a system immediately regains its prestress capacity. Hysteresis and memory may delay the expansion of risk until stability has persisted long enough. Entry into and exit from defense therefore use different conditions in the capital controller, and exposure should increase more slowly than it decreases.

The value of allostasis in this study is not to attach biological labels to financial indicators, but to supply an integrated control logic that combines anticipatory preparation, multi-speed response, accumulated cost, and asymmetric recovery.

1.4. Research Gap and Competing Explanations

Physiology, resilience engineering, probabilistic forecasting, drawdown control, and dynamic portfolio management already contain related components. The substantive gap is their rigorous integration into a falsifiable mechanism. Existing evidence does not establish whether a complete chain from physiological theory to function, abstraction, artificial state, control action, and deletion test can improve capital viability in nonstationary financial environments. Nor is it clear whether a composite allostatic load adds information beyond volatility and drawdown; whether fast-slow loops are merely multiple moving windows; whether recovery memory exceeds an ordinary cooldown rule; or whether dynamic risk boundaries outperform generic prediction intervals.

Parameter count is another important competing explanation. A full controller may appear superior only because it is more flexible. A credible design must therefore compare the proposed mechanism with a nonlinear, complexity-matched nonbiological model, constant-exposure benchmarks, mechanism ablations, and a reversed mechanism. If these alternatives perform equally well or better, performance differences cannot be attributed to the proposed biological mechanism.

XCMG Machinery provides a demanding longitudinal validation environment: approximately three decades of daily observations, multiple corporate and market regimes, 91 independent drawdown cycles, and several complete stress-recovery processes. This time span supports within-case mechanism process tracing, but it does not create multiple independent securities, eliminate firm-specific history, or establish external validity. Daily observations are serially dependent, and thirty calendar years do not imply an invariant data-generating mechanism.

The central research question is therefore: Can anticipatory, multi-timescale, load-accumulating, and recovery-memory functions translated from physiological allostasis improve risk-boundary maintenance and shock recovery in an artificial capital process exposed to a changing financial environment?

1.5. Research Objective and Falsifiable Hypotheses

This study develops the Biomimetic Allostatic Risk and Viability Controller (BARVC). The controller receives conditional risk forecasts and states representing fast stress, slow structure, accumulated load, and recovery memory, and it outputs bounded risk exposure and asymmetric re-entry rules. Five prespecified hypotheses are tested:

H1—Anticipatory regulation. Compared with a purely reactive controller, the full controller will produce fewer severe breaches and achieve faster recovery.

H2—Multiple timescales. A dual fast-slow loop will produce better out-of-sample calibration and risk-control performance than a single-timescale controller.

H3—Incremental validity of load. After conventional volatility, drawdown, duration, and trend variables are included, allostatic load will still improve probabilistic scoring and calibration.

H4—Recovery memory. Relative to a memory-free model, hysteretic recovery will reduce the risk of premature expansion and secondary shocks.

H5—Mechanistic integrity. Deleting any major mechanism will worsen at least one prespecified primary endpoint without causing failure on another primary endpoint.

If these conditions are not met, the corresponding hypothesis is classified as unsupported. Marginal differences, post hoc graphical plausibility, or superiority only to a weak benchmark do not constitute support.

1.6. Contributions and Scope

The intended biological contribution is a transparent translation of allostatic functions into computational modules that can be deleted or reversed. The methodological contribution is an integrated architecture connecting conditional quantiles, fast and slow states, accumulated load, hysteretic recovery, and a virtual capital process. The empirical contribution is a long-horizon within-case test across multiple stress-recovery cycles. The evaluative contribution is a shift from directional accuracy toward coverage error, severe risk breaches, maximum drawdown, time underwater, recovery time, turnover, and capital preservation.

All of these contributions are conditional on forward validation and discrimination from simpler alternatives. The study tests the computational feasibility of functional translation; it does not establish biological homology, a general theory of stock prediction, a stable law of excess returns, or cross-asset applicability.

2. Biological Foundations and Mechanism Translation

2.1. From Regulated Variables to Viability

Classical homeostatic frameworks center on regulated variables and negative feedback [35,36]. Allostasis broadens the unit of analysis: regulation is organized around anticipated demand and organism-level viability, which may require coordinated changes across several subsystems [37-39]. The key design translation is not the replication of a physiological molecule but the translation of a control relationship in which environmental cues and internal states alter resource allocation before, during, and after a disturbance.

BARVC treats virtual capital as the viability process and risk exposure as the principal actuator. The controller does not repair a security's price; it changes the proportion of external returns transmitted to the capital path. The P10-P90 interval can therefore be interpreted as a viability boundary only when it participates in control decisions. If it merely displays a future price distribution, it remains a conditional prediction interval.

Figure 1 presents the functional chain through which environmental signals enter anticipatory regulation, fast-slow loops generate allostatic load and recovery memory, and these states ultimately govern risk exposure, the risk budget, and re-entry conditions. The figure also marks feedback relations and the theoretical boundary between functional correspondence and biological homology.

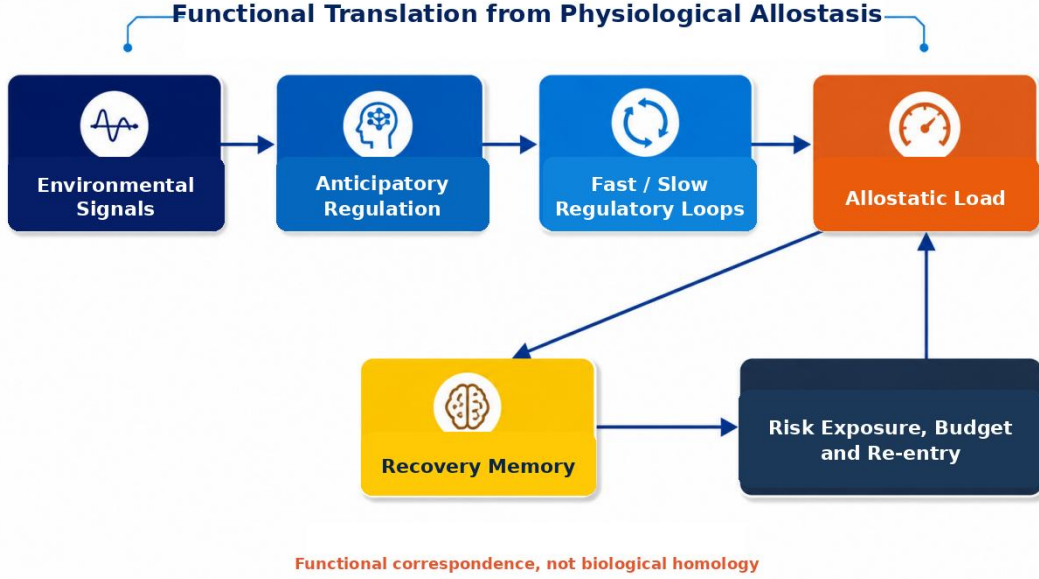


Figure 1. Functional translation from physiological allostasis to artificial risk regulation. The mapping is functional rather than biologically homologous.

2.2. Anticipatory Regulation

Anticipatory regulation uses imperfect cues to prepare resources before demand peaks [38,39]. BARVC maps this principle to a lower-tail forecast of the 20-day return distribution. When the predicted lower quantile deteriorates, the system can reduce risk exposure before realized drawdown crosses a feedback threshold. Because forecasts can fail, prediction enters the control score only as a weighted component. Exposure is constrained to $[0, I]$, and leverage is prohibited.

The falsification path is direct: deleting the predictive component yields a purely reactive model. If the full model does not reduce severe-breach frequency or improve recovery relative to that model, H1 is unsupported. This design separates the test of a biological function from the claim that the forecast itself is accurate.

2.3. Fast and Slow Regulation

The fast state is an exponentially weighted response to acute volatility, price range, drawdown, duration, and breach stimuli:

$$F_t = \alpha F_{t-1} + (1 - \alpha) \phi(x_t), \quad \alpha = 0.80.$$

The slow state integrates the same environment over a longer memory scale:

$$S_t = \beta S_{t-1} + (1 - \beta) \psi(x_{t-h:t}), \quad \beta = 0.98.$$

The corresponding financial half-lives are approximately 3 and 34 trading days. These values are computational instances of fast and slow behavior, not quantities derived directly from human physiology. Single-loop and loop-deletion models assess whether the combination provides information beyond that of conventional multi-window smoothing.

2.4. Allostatic Load

Allostatic load is represented as a bounded, persistent state:

$$L_t = \rho L_{t-1} + (1 - \rho)(w_1 V_t + w_2 D_t + w_3 T_t + w_4 Q_t + w_5 R_t), \quad \rho = 0.97,$$

where V_t denotes volatility stress, D_t drawdown depth, T_t the log-transformed duration underwater, Q_t the recent frequency of lower-tail breaches, R_t recurrent stress. Inputs are converted to empirical percentile ranks from the training period and bounded to $[0, 1]$; weights are nonnegative and sum to 1. Persistence prevents a single calm observation from erasing a long history of stress.

This composite may still be no more than a relabeling of its components. Incremental validity is supported only if it continues to improve performance after conventional variables have entered the model. If calibration or proper scoring does not improve, the load remains computationally definable but lacks independent empirical value in this case.

2.5. Recovery Memory and Hysteresis

Recovery memory H_t combines recent drawdown, duration, load, and failed stabilization and decays slowly enough to constrain risk re-entry. BARVC enters a defensive state when the composite score reaches 0.48; exposure is capped at 0.40 during defense; and defense ends only after the score remains at or below 0.28 for 10 consecutive trading days. Exposure can contract immediately, but its daily increase is capped at 0.03.

This asymmetry implements recovery hysteresis. A memory-free ablation and a fixed-threshold controller test whether it adds value beyond a generic cooldown rule. The reversed-mechanism controller increases exposure as stress rises and accelerates recovery under high load to test directional validity; interpretation must nevertheless account for differences in mean exposure.

2.6. Biology-to-Algorithm Translation Matrix

Table 1 maps each biological source sequentially to its physiological function, abstract principle, artificial computational module, control output, and falsification test. The matrix allows later ablations to specify rejection conditions for particular mechanisms rather than infer biomimetic status from an algorithm’s name.

Table 1. Functional translation and falsification design for physiological allostatic mechanisms and artificial risk-regulation modules

Biological source	Physiological function	Abstract principle	Artificial state/module	Control output	Falsification test
Anticipatory regulation	Prepare resources before demand	Feedforward regulation	Conditional lower-tail forecast	Contract the risk budget in advance	Delete prediction; compare with the reactive model
Fast response	Handle acute disturbances	Short-memory feedback	F_t	Emergency defense	Delete the fast loop
Slow response	Reset baselines and sensitivity	Long-memory coordination	S_t	Structural budget adjustment	Delete the slow loop or retain one loop only
Allostatic load	Record cumulative regulatory cost	Persistent stress state	L_t	Reduce capacity and delay re-entry	Delete load and test incremental validity
Recovery memory	Prevent premature reset	Hysteresis and path dependence	H_t	Asymmetric exit/re-entry	Delete memory and measure secondary shocks

The matrix specifies what would count as evidence for a biomimetic mechanism: not a favorable-looking graph, but a localized improvement that persists after a specific module is deleted or reversed and after comparison with complexity-matched controls.

3. Methods and Research Design

3.1. Research Design and Study Asset

This study uses a single-case, long-time-series, longitudinal proof-of-concept design focused on mechanism translation. The study security is XCMG Machinery (Shenzhen Stock Exchange: 000425), with daily records from August 28, 1996, through August 12, 2026. Returns, technical states, and drawdowns use forward-adjusted open, high, low, and close prices. Unadjusted prices are used only for contemporaneous price interpretation and are not included in the controller experiment.

XCMG Machinery is the validation environment, not a statistical population. The locked dataset contains no market index, industry panel, delisted-firm panel, or international asset sample; market-wide, industry-specific, and firm-specific shocks therefore cannot be separately identified.

3.2. Data-Integrity Audit

The raw daily file contains 7,109 records and 7,109 unique dates. Open, high, low, close, volume, turnover value, and turnover rate contain no missing observations. The audit found no negative volume or turnover value, zero-volume trading days, duplicate dates, or OHLC logic violations: the high is never below another price field, and the low is never above another price field. All records appear in ascending chronological order. We retained extreme returns because the adjusted series showed no mechanical discontinuity that met the prespecified data-error rule.

The data include three price bases and a separate adjustment-factor file. All model calculations consistently use forward-adjusted prices to avoid misclassifying corporate actions as return shocks. We may disclose derived data and code, whereas the original provider’s license and terms govern the redistribution of raw market data.

3.3. Drawdown and Upward-Move Events

Daily drawdown is defined as:

$$DD_t = \frac{P_t}{\max_{s \leq t} P_s} - 1.$$

An independent drawdown cycle begins after a historical closing-price high and ends when the price first returns to or exceeds that high. Multiple local troughs occurring before recovery are assigned to the same cycle. Thresholds of 10%, 20%, 30%, and 50% are used descriptively. Full capital recovery means that price returns to the previous high; functional recovery under BARVC additionally requires lower load and 10 consecutive trading days of state stability.

The upward-move algorithm maintains a running local low. An event is activated after price rises 15% from that low, continues tracking the peak, and closes after an 8% decline from the peak. The procedure identifies 124 events. Because each starting point is a local low identified retrospectively, event frequency cannot be interpreted as an ex ante hit rate.

3.4. Temporal Splits and Leakage Control

The training period is 1996-2018, the calibration period is 2019-2021, and the locked test period is 2022-2026. The primary forecast horizon is 20 trading days. Horizons of 5, 60, 120, and 252 trading days were prespecified only for future extensions and were not substituted for an underperforming primary horizon. The purge rule removes labels whose future windows cross a training or calibration boundary, reducing leakage from overlapping labels and time-series validation [45,46]. The final analytical sample contains 5,266 training observations, 697 calibration observations, and 1,083 locked test observations. Because a complete 20-day future outcome is required, the final label-eligible test origin is July 15, 2026.

All rolling variables use only information available at or before time t . Median imputation, robust scaling, empirical percentile-rank mapping, and model fitting use training-period information only; calibration residuals are estimated from 2019-2021. We excluded all locked-test observations from variable selection, weight estimation, threshold setting, and cost-assumption specification.

3.5. Perceptual Variables

Inputs include 5-, 20-, 60-, 120-, and 252-day returns; 20- and 60-day annualized realized volatility; the 14-day average true range-price ratio; current high-water-mark drawdown; log time underwater; abnormal volume; 5/20-day volume and turnover-rate ratios; distance from 20-, 60-, 120-, and 252-day moving averages; Bollinger-band z score; 20/120-day trend strength; Wilder’s RSI; and recent lower-tail breach frequency.

The first experiment excludes news, corporate disclosures, accounting forecasts, analyst recommendations, order-book data, and sentiment measures. This restriction reduces flexibility in feature search and keeps the study focused on the translated regulatory mechanisms.

3.6. Conditional Quantile Models

The prediction target is the 20-day forward return $Y_{t+20} = P_{t+20}/P_t - 1$. Three model classes are evaluated: (1) historical unconditional P10, P50, and P90; (2) a non-allostatic gradient-boosting quantile model using conventional perceptual variables; and (3) the same model augmented with F_t, S_t, L_t, H_t . Each quantile model uses 220 trees of maximum depth 2, a learning rate of 0.03, and a minimum of 40 samples per leaf. Hyperparameters were fixed before the locked test.

Calibration-period residual quantiles are used for additive correction. If predicted quantiles cross, they are reordered monotonically. The dynamic boundary is:

$$[B_t^L, B_t^U] = Q_{0.10, 0.90}(Y_{t+20} | F_t, S_t, L_t, H_t, M_t).$$

Coverage error is defined as $|\hat{C}_{0.80} - 0.80|$, and sharpness is measured by mean interval width. Quantile accuracy is assessed with pinball loss:

$$\mathcal{L}_q(y, \hat{y}) = \max\{q(y - \hat{y}), (q - 1)(y - \hat{y})\}.$$

The mean of the three quantile losses is reported as a sparse-quantile score. Because only three quantiles are estimated, this measure is not described as a full continuous ranked probability score, which requires evaluation over a more complete predictive distribution [47].

3.7. BARVC Actuation Function

Training-period empirical percentile ranks map risk stimuli to $[0, 1]$. The prespecified full score is:

$$Z_t = 0.22F_t + 0.18S_t + 0.25L_t + 0.15H_t + 0.20\widehat{R}_t,$$

Here, $\widehat{R}_t$ converts the predicted P10 return into a bounded lower-tail stress measure. The unconstrained target exposure is $1 - Z_t$, after which defensive and overload caps, hysteresis rules, and the 0.03 daily recovery increment are applied:

$$e_t = \text{clip}\{g(F_t, S_t, L_t, H_t, \widehat{Q}_{t+20}), 0, 1\}.$$

Virtual-capital return is:

$$r_{t+1}^p = e_t r_{t+1} - c|e_t - e_{t-1}|.$$

The cost per unit of change in one-way exposure is 15 basis points: 10 basis points for transaction costs and 5 basis points for slippage. Leverage is prohibited so that apparent performance cannot be manufactured through implicit borrowing.

3.8. Benchmarks and Ablation Models

Benchmarks include buy-and-hold, constant 45% exposure, a 20% volatility target, fixed-drawdown control, quantile-only control, and a gradient-boosting, complexity-matched nonbiological stress classifier. Constant 45% exposure approximates the controller's ex ante risk scale and therefore provides a demanding parsimonious benchmark.

Ablations separately remove anticipatory regulation, the fast state, the slow state, load, or recovery memory. The single-timescale model retains only the fast loop. The reversed-mechanism model makes risk exposure increase with the raw stress score. All variants use the same data and cost rate. Mean exposure is reported for the reversed model to avoid mistaking a simple reduction in risk exposure for directional validation.

3.9. Endpoints and Statistical Analysis

The primary endpoints are (1) absolute P10-P90 coverage error; (2) the proportion of locked-test days on which virtual-capital drawdown is at least 20%; and (3) recovery duration after capital drawdown reaches 10%. Secondary endpoints include interval width, pinball loss, maximum drawdown, proportion of time underwater, annualized volatility, terminal wealth, turnover, the number of recovery events, and median recovery days.

Because observations overlap and are serially dependent, daily values are not treated as independent. A 20-day moving-block bootstrap with 1,000 repetitions estimates confidence intervals for paired differences in coverage and severe-breach rates [48,49]. Inference for predictive-loss comparisons and risk-adjusted measures must likewise account for serial dependence [50,51]. Each controller produces only 4-8 recovery events in the locked test period, so event outcomes are interpreted descriptively. Multiple hypotheses are judged by prespecified directional rules, not by isolated p values obtained by treating daily observations as independent. Future multi-asset studies should add event-block inference, model confidence sets [52], conditional-coverage tests [19], and multiple-comparison corrections [53,54].

3.10. Reproducibility and Audit Trail

The analysis fixes the random seed at 425. The code preserves the data-audit summary, state sequences, test-period quantiles, risk exposures, drawdown cycles, controller metrics, and figures. We retained all prespecified hypotheses, including those that the results did not support, and we did not retune the full controller after observing the locked-test results. This distinction is essential: the experiment evaluates a mechanism specified in advance rather than a controller optimized retrospectively on the same historical data.

4. Results

4.1. Data and Long-Term Stress Characteristics

The audit confirms 7,109 valid trading days from August 28, 1996, through August 12, 2026, and identifies 91 independent high-water-mark drawdown cycles. Twelve exceed 20%, seven exceed 30%, and four exceed 50%. The maximum-drawdown cycle began from a forward-adjusted closing-price peak of CNY 1.74 on June 24, 2002, reached a trough of CNY 0.18 on July 18, 2005, and recovered on February 15, 2007. The maximum drawdown was 89.66%; peak-to-trough decline lasted 733 trading days, and total time underwater was 1,082 trading days. Recovery from the trough to the prior high required an 866.7% gain.

The second-deepest drawdown cycle began in March 2011, reached 79.4%, and did not recover to its prior high until February 2025, remaining underwater for 3,344 trading days. These events demonstrate that drawdown depth and recovery duration capture different dimensions of downside risk and should therefore be assessed separately. Figure 2 places thirty years of prices and synchronous high-water-mark drawdowns on a common timeline to reveal shifts in volatility regimes and prolonged deviations from historical highs. The upper panel uses a logarithmic price axis, and the lower panel marks a 20% stress reference line, thereby avoiding visual distortion from nominal price growth alone.

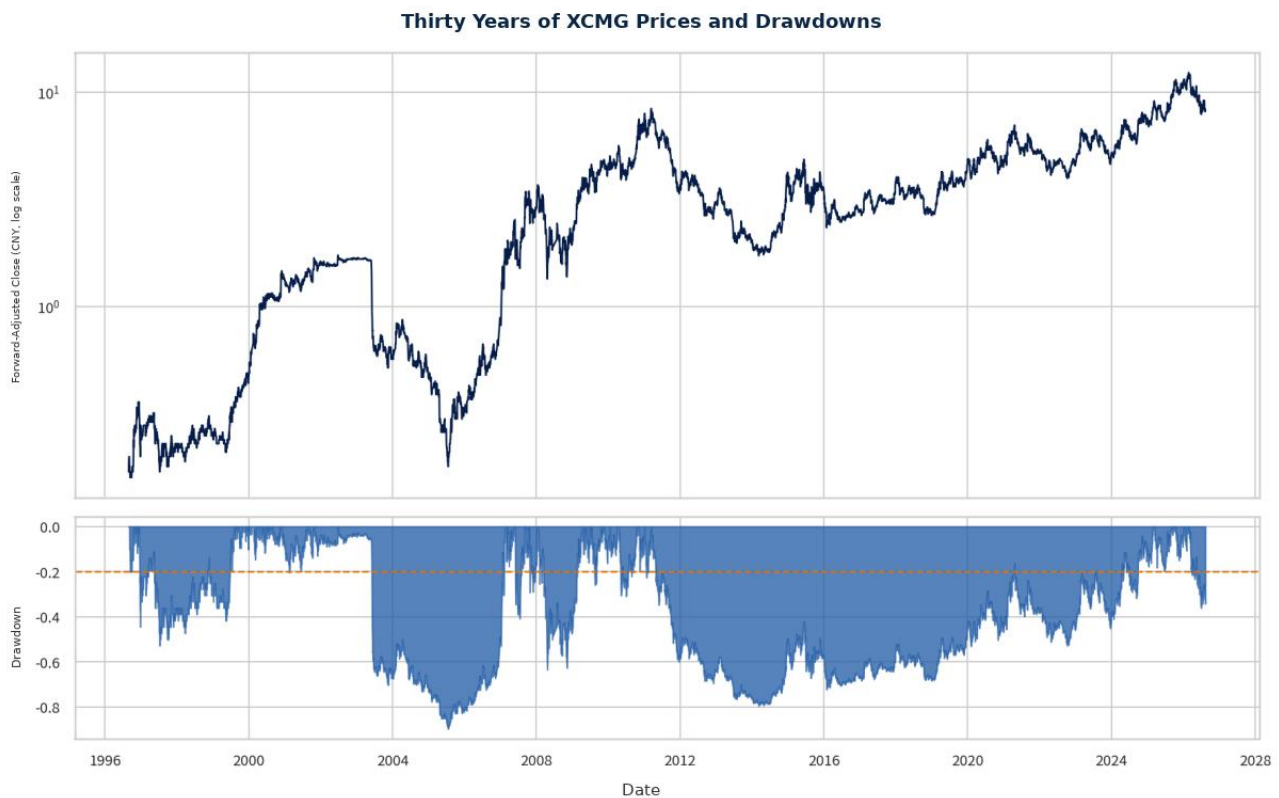


Figure 2. Thirty years of forward-adjusted XCMG closing prices and high-water-mark drawdowns.

Table 2 summarizes the sample size, number of independent drawdowns, stratified stress events, and historical maximum-drawdown cycle corresponding to Figure 2. It supplies auditable exact values and reports peak-to-trough loss separately from time underwater.

Table 2. Descriptive statistics for the thirty-year sample and independent drawdown cycles

Statistic	Value
Valid trading days	7,109

Statistic	Value
Independent drawdown cycles	91
Drawdown $\geq 20\%$	12
Drawdown $\geq 30\%$	7
Drawdown $\geq 50\%$	4
Historical maximum drawdown	89.66%
Maximum-drawdown peak/trough/recovery dates	2002-06-24 / 2005-07-18 / 2007-02-15
Peak-to-trough/total time underwater	733 / 1,082 trading days

4.2. Retrospective Description of Technical Features Before Upward Moves

The 15% rise/8% reversal event algorithm reproduces 124 major upward-move events. Median event return is 27.8%, and the median low-to-peak duration is approximately 22 trading days. At the retrospectively identified lows, 87.9% of events begin below the 20-day moving average, 90.3% occur during a 60-day drawdown greater than 10%, 60.5% have a Wilder RSI below 40, and only 22.6% have a 5/20-day volume ratio above 1.10. The median distance from the 20-day moving average is -6.3% , the median 60-day drawdown is -19.0% , and the median RSI is 36.57.

These observations support a descriptive post-stress-repair pattern but do not constitute a forward trading rule. In a previously locked classification experiment using the same data, the 20-day upward-event classifier achieved a ROC AUC of only 0.469 and average precision of 0.200, against a positive-class base rate of 0.208; the highest quintile of predicted probabilities achieved a hit rate of only 15.5%. This negative result is retained because it directly limits interpretation of the 124 events.

4.3. Allostatic State Trajectories

The fast state reacts sharply to short-run shocks; the slow state and load retain stress over longer intervals; and recovery memory is smoother and remains elevated after prolonged underwater periods. Figure 3 displays the four states on a common 0-1 normalized scale and shades the locked test period, permitting direct comparison of response speed and persistence. Because the states share upstream stimuli, they are correlated. The figure demonstrates computational separation, not the empirical necessity of every state.

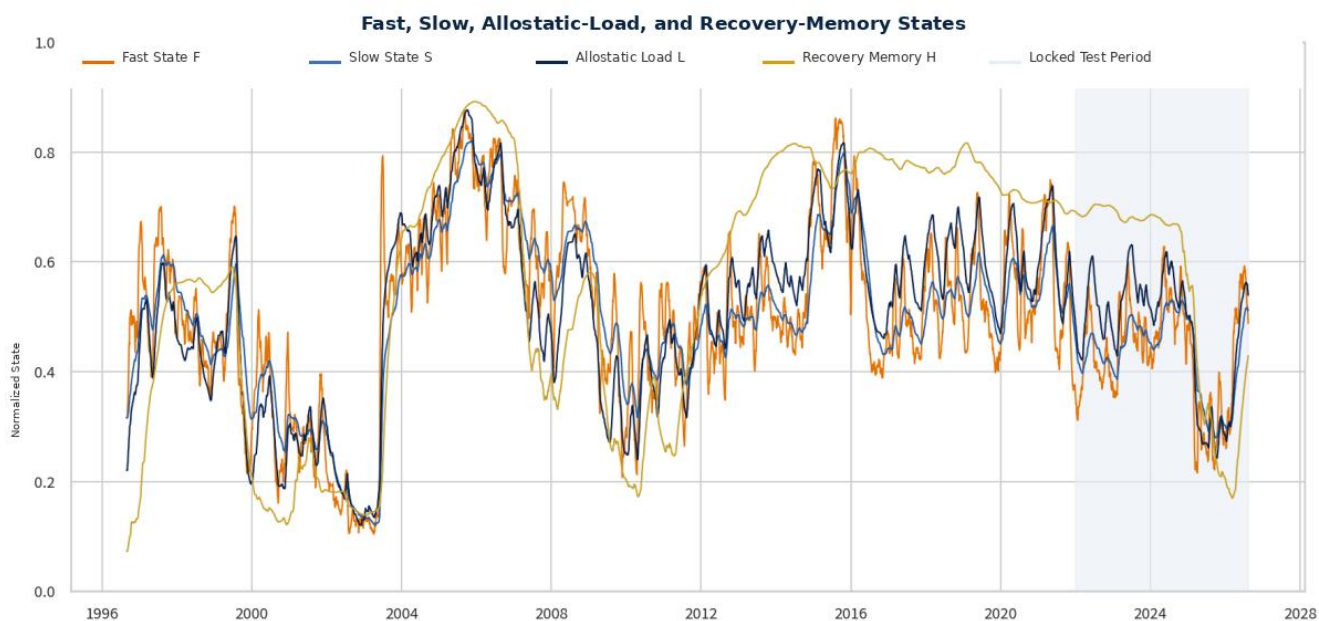


Figure 3. Normalized fast state, slow state, allostastic load, and recovery memory. The shaded area denotes the locked test period.

4.4. Conditional Quantile Performance

The historical unconditional interval covered 94.92% of locked-test outcomes, well above the nominal 80% level, and had a mean return-interval width of 33.16 percentage points. It was conservative but unsharp. The non-allostastic conditional model reduced width to 24.42 points, achieved 75.35% coverage, and produced an absolute coverage error of 4.65 points. Adding allostastic states narrowed the interval further to 22.50 points but reduced coverage to 72.02%, increasing coverage error to 7.98 points. Mean pinball loss improved slightly from 0.02479 to 0.02441, but calibration deteriorated materially.

The paired 20-day block bootstrap estimates the coverage difference—full BARVC minus the non-allostastic model—at -3.32 percentage points, with a 95% confidence interval of $[-6.09, -1.29]$ points. H3 is therefore unsupported: the augmented states did not jointly improve calibration and scoring.

Table 3 reports coverage, absolute coverage error, mean width, and mean pinball loss for the three probabilistic models. Reading these measures jointly prevents high coverage generated by an excessively wide interval from being misclassified as better probabilistic forecasting.

Table 3. Calibration, sharpness, and loss of conditional quantile models in the locked test period

Model	P10-P90 coverage	Absolute coverage error	Mean width	Mean pinball loss
Historical unconditional model	94.92%	14.92 percentage points	0.3316	0.02419
Non-allostastic conditional model	75.35%	4.65 percentage points	0.2442	0.02479
Full-state BARVC	72.02%	7.98 percentage points	0.2250	0.02441

Figure 4 presents the full-state model's daily P10-P90 price boundary, P50 path, and realized 20-day-ahead price. Breaches of the upper and lower boundaries cluster over time, suggesting that undercoverage reflects conditional-calibration failure rather than independent random misses.

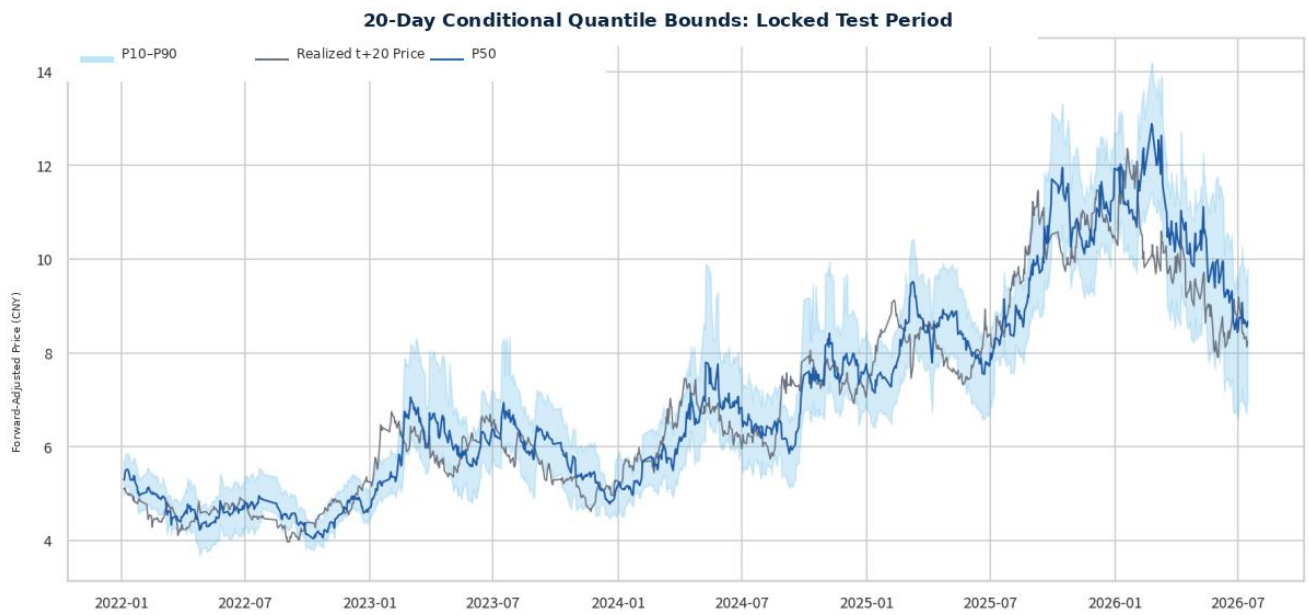


Figure 4. Realized 20-day-ahead price and full-state P10-P90 conditional interval in the locked test period.

Figure 5 compares the nominal 80% coverage target with realized coverage for all three models and simultaneously displays mean pinball loss. The comparison shows that modest gains in interval sharpness and quantile loss do not offset the calibration cost represented by the marked decline in coverage.

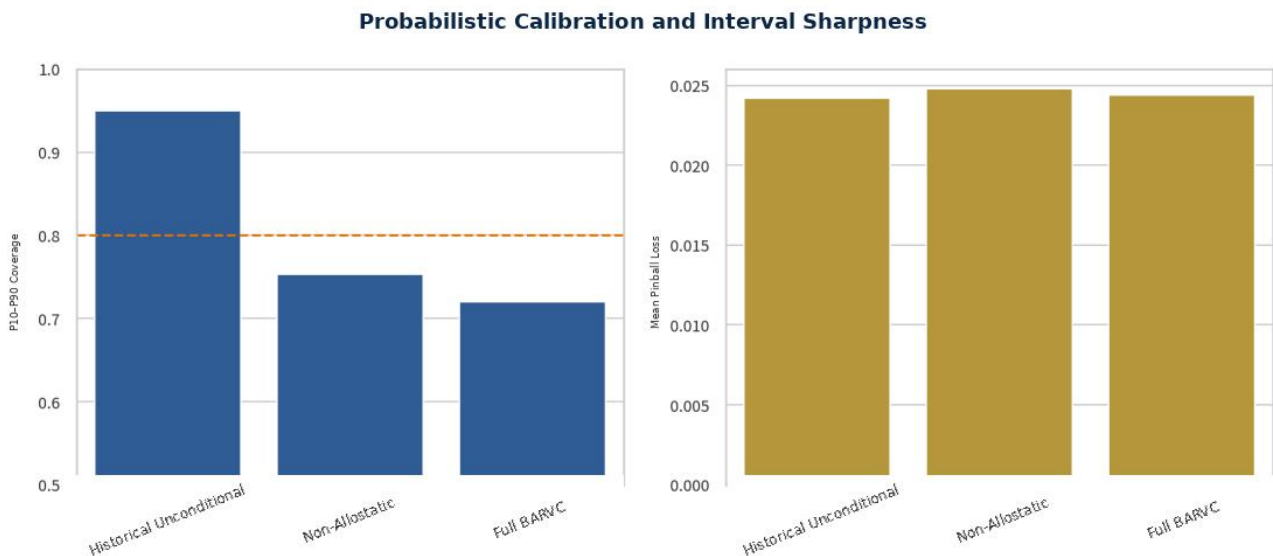


Figure 5. Coverage and mean pinball loss for the historical, non-allostatic, and full-state models.

4.5. Virtual-Capital Performance

At the end of the labeled locked-test period, buy-and-hold produced terminal wealth of 1.617, an annualized return of 11.82%, annualized volatility of 33.80%, maximum drawdown of 36.08%, and capital drawdown of at least 20% on 16.16% of trading days. BARVC had mean exposure of 44.96% and terminal wealth of 1.219; annualized volatility declined to 16.01%, maximum drawdown to 21.77%, and the severe-breach rate to 0.83%. The bootstrap interval for the severe-breach-rate difference between the full model and buy-and-hold was $[-22.53, -7.48]$ percentage points.

That comparison alone, however, would overstate the mechanistic effect. Constant 45% exposure, which approximates BARVC's risk budget, produced terminal wealth of 1.318, volatility of 15.21%, maximum

drawdown of 17.41%, no severe breaches, and no turnover costs. It therefore outperformed BARVC on terminal wealth, volatility, maximum drawdown, severe breaches, and turnover. The complexity-matched classifier had mean exposure of 85.41%, terminal wealth of 1.587, and maximum drawdown of 30.81%; its higher return and higher risk mainly reflect a different exposure level and do not constitute a clean mechanism comparison.

Table 4 summarizes risk exposure, terminal wealth, return, volatility, drawdown, severe breaches, and turnover for eight controllers under the same locked test period and cost assumptions. A constant 45% exposure is the critical benchmark for interpreting the source of BARVC 's risk reduction, as it has an approximately equivalent mean risk budget.

Table 4. Risk, capital, and transaction-cost-related metrics for controllers in the locked test period

Controller	Mean exposure	Terminal wealth	Annualized return	Annualized volatility	Maximum drawdown	Severe-breach days	Annualized turnover
Buy-and-hold	100.00%	1.617	11.82%	33.80%	36.08%	16.16%	0.000
Constant 45% exposure	45.00%	1.318	6.64%	15.21%	17.41%	0.00%	0.000
Volatility target	67.49%	1.349	7.22%	22.14%	27.16%	6.46%	5.893
Fixed-drawdown control	49.39%	1.048	1.10%	19.70%	30.77%	5.63%	4.002
Quantile-only control	64.09%	1.565	10.98%	21.18%	24.58%	2.12%	4.442
Complexity-matched model	85.41%	1.587	11.34%	28.54%	30.81%	9.51%	2.251
Purely reactive model	44.74%	1.235	5.04%	16.06%	21.26%	0.65%	0.246
Full BARVC	44.96%	1.219	4.71%	16.01%	21.77%	0.83%	0.343

Figure 6 displays virtual-capital paths for buy-and-hold, the complexity-matched model, and full BARVC, with BARVC risk exposure shown below. The figure is used to identify synchronization between capital-path differences and exposure contraction, not to interpret the mechanically lower drawdown arising from lower exposure as adaptive capacity.

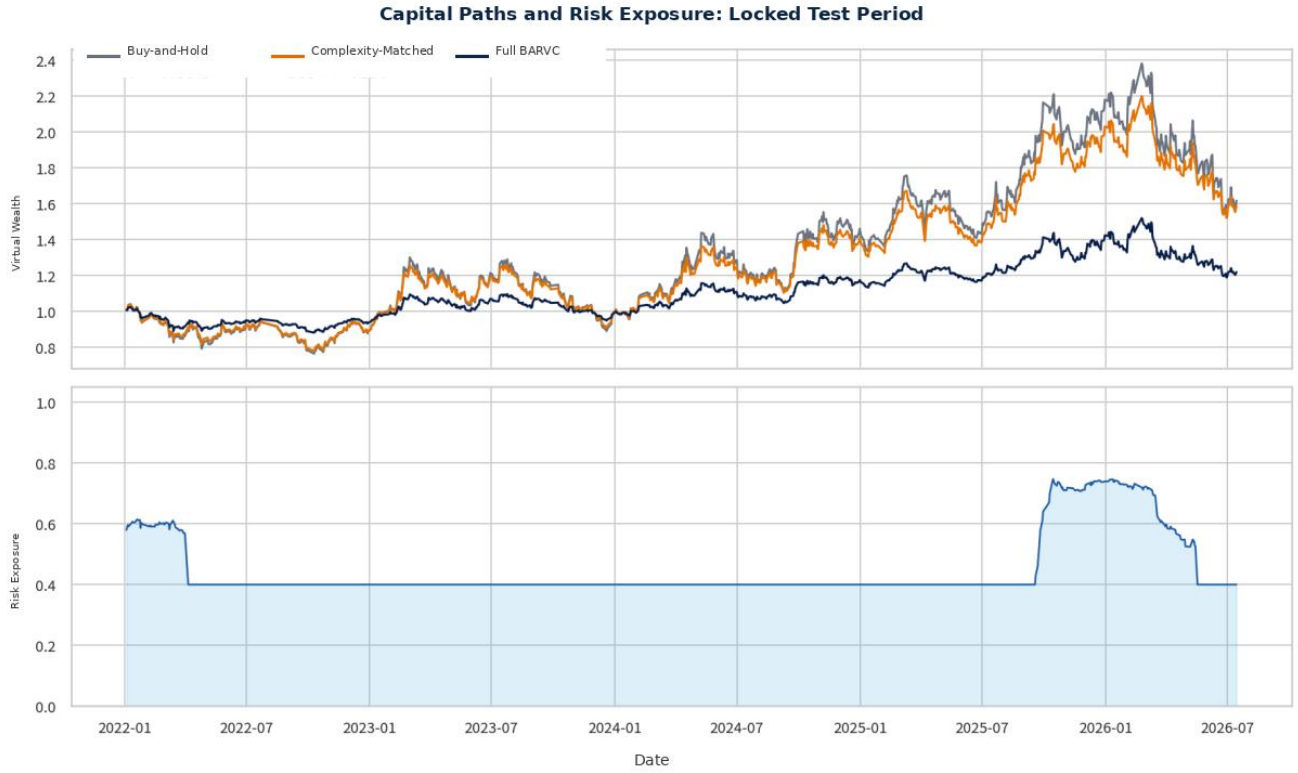


Figure 6. Virtual-capital paths and BARVC risk exposure in the locked test period.

Figure 7 displays maximum-drawdown magnitude and terminal wealth for the eight controllers in paired panels so that risk protection and capital growth can be assessed jointly. The figure specifically tests whether BARVC outperforms a simpler constant-exposure benchmark at a comparable risk budget.

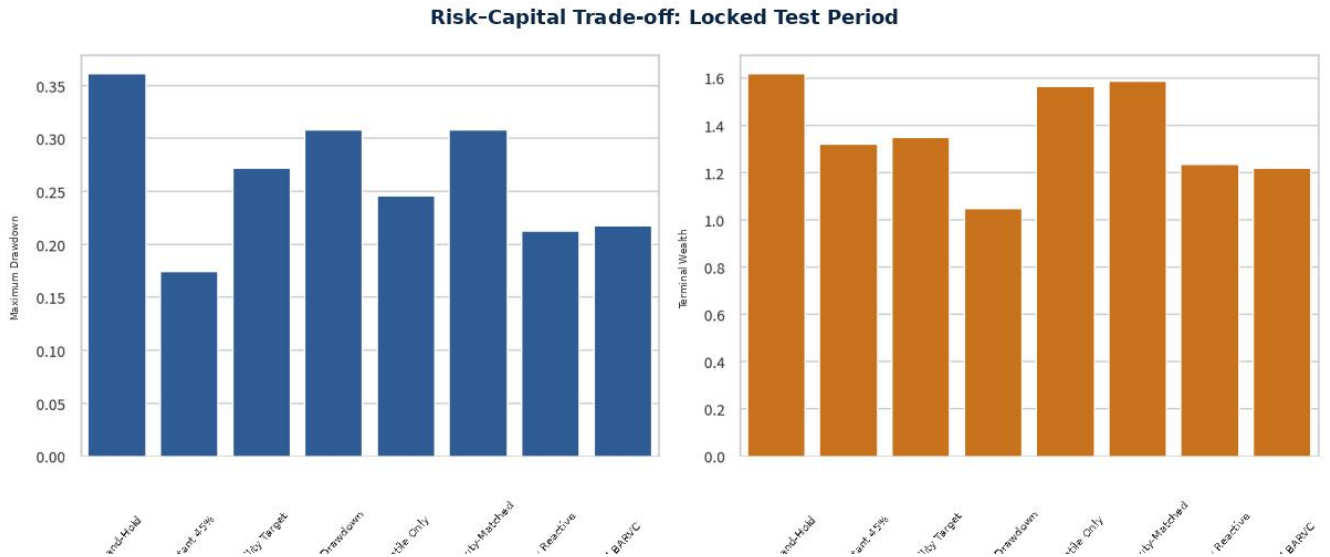


Figure 7. Comparison of maximum drawdown and terminal wealth.

4.6. Recovery and Ablation Experiments

At a 10% capital drawdown threshold, buy-and-hold produced eight events, seven of which recovered within the test period, with a median recovery time of 83 trading days. BARVC produced four events, three recoveries, and a median recovery of 84 trading days. Because event counts are small and differ across controllers, a formal claim of recovery-time advantage is not warranted. Constant 45% exposure likewise produced four events and

three recoveries, with a median recovery of 107 trading days. Event frequency and recovery duration thus capture different aspects of resilience.

The full model did not outperform the purely reactive, no-slow-state, no-load, or no-memory models in terms of maximum drawdown. The no-memory model had a 20.74% maximum drawdown and a 0.37% severe-breach rate; the purely reactive model produced 21.26% and 0.65%, respectively. Removing the fast loop worsened the maximum drawdown to 22.32% and increased severe breaches to 1.48%, but this localized deterioration is insufficient to support mechanistic integrity, as other deletions improved primary endpoints.

The reversed-mechanism model had mean exposure of only 36.41%, so its performance cannot be treated as a risk-budget-equivalent comparison. Its maximum drawdown was 13.69%, and its severe-breach rate was zero, showing that the current test did not establish the unique validity of the proposed direction. Future studies must match reversed controllers on exposure or volatility ex ante.

Figure 8 presents the maximum drawdown, severe breaches, and turnover for the full model, major mechanism-deletion models, the single-scale model, and the reversed-mechanism model in a single matrix. Color is used only for within-column comparison, and each cell also displays the raw value so that the normalized scale does not conceal the actual magnitude of an indicator.

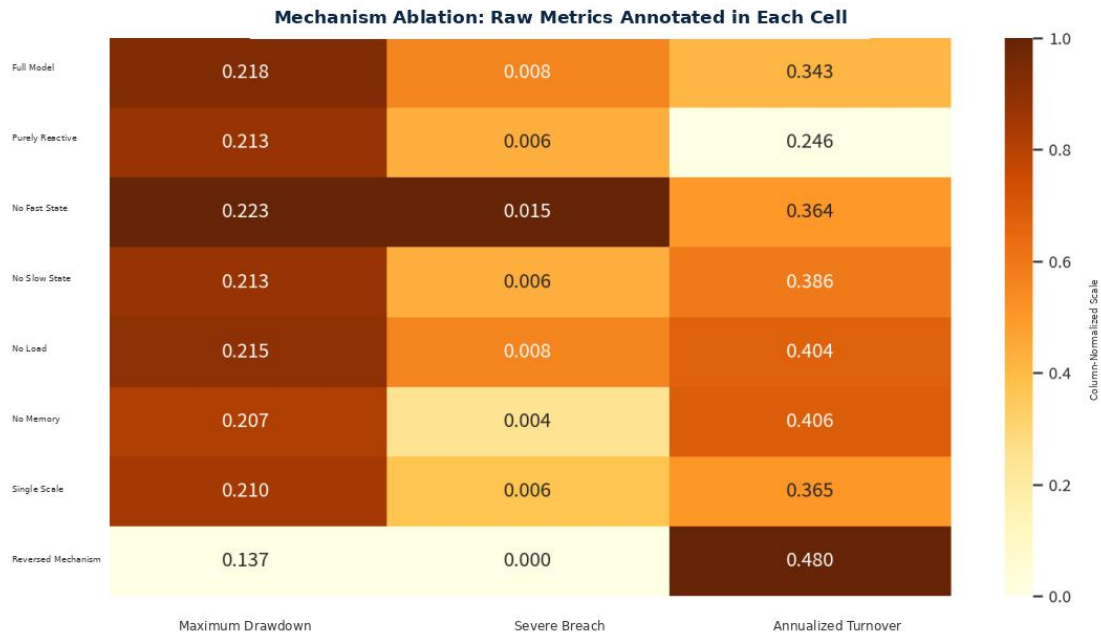


Figure 8. Ablation results. Each cell displays the raw metric; color is normalized by column.

Table 5 records the observed results and determinations for each of the five mechanism hypotheses under the prespecified criteria. It separates observed numerical differences from the question of whether the support condition was met, preventing a local improvement from being mistaken for the complete hypothesis test.

Table 5. Results and determinations for the prespecified mechanism hypotheses

Hypothesis	Prespecified criterion	Observed result	Determination
H1 Anticipatory regulation	Full model improves breaches and recovery relative to the reactive model	Full: 0.83% breaches, median 84 days; reactive: 0.65%, 84 days	Unsupported
H2 Multiple timescales	Calibration and risk outperform the single-timescale model	Coverage deteriorated; full-model drawdown 21.77%, single-scale 20.98%	Unsupported
H3 Incremental validity of load	Scoring improves without worse calibration	Pinball loss improved slightly; coverage error worsened by 3.33 percentage points	Unsupported
H4 Recovery memory	Reduce premature expansion or secondary shocks	The no-memory model had lower drawdown and breach rates	Unsupported

Hypothesis	Prespecified criterion	Observed result	Determination
H5 Mechanistic integrity	Deleting any module worsens a primary endpoint	Several deletions improved primary endpoints	Unsupported

4.7. Robustness Interpretation

The most robust finding is not BARVC superiority but the need to distinguish a lower-risk budget from genuine adaptive capacity. Lower mean exposure mechanically reduces the volatility and drawdown transmitted to the capital process. Once a constant-exposure benchmark is introduced, most of the apparent advantages of buy-and-hold disappear. This is an important negative control because many dynamic strategies receive credit for risk control that a lower-cost static de-risking policy could reproduce.

The probabilistic results are consistent with the negative mechanistic determinations. Allostatic states narrowed the intervals but led to undercoverage; the historical interval was overcovered because the earlier sample exhibited a wider stress distribution. These results are consistent with structural change: calibration relationships from 2019-2021 did not transfer adequately to 2022-2026.

5. Discussion

5.1. Principal Findings

The study successfully translates physiological allostatic functions into an auditable artificial controller, but it does not validate the superiority of the complete mechanism. BARVC substantially reduces capital-path volatility and drawdown relative to buy-and-hold, yet a constant risk budget with almost identical mean exposure provides stronger protection and higher terminal wealth. None of the five prespecified mechanism hypotheses is supported. The correct conclusion is therefore that computational translation is feasible but remains empirically unconfirmed in one locked case.

This result has scientific value because it distinguishes three propositions that are often conflated: a biological theory may be conceptually coherent; researchers may be able to implement a computational analog of that theory; yet the resulting analog may still fail to provide incremental predictive or control value in a particular environment. Credible biomimetic research therefore requires separate tests of conceptual coherence, computational implementability, and incremental predictive or control value.

5.2. Interpretation of the Biological Mechanism Translation

The state trajectories qualitatively conform to their design: the fast loop changes quickly, the slow loop and load are more persistent, and recovery memory decays gradually. These are implementation properties, not evidence that financial and physiological stress share the same material basis. Functional correspondence remains intact: cues support preparation, different response speeds change budgets, historical stress accumulates, and risk re-entry is asymmetric.

The mechanisms may have failed to outperform for four reasons. First, the states are constructed from overlapping price variables and are highly redundant with conventional features; allostatic load may add persistence without adding new information. Second, one security supplies only a small number of mutually independent extreme events. The 7,109 trading days are not 7,109 independent regulatory experiments; for severe recovery questions, the effective sample size is closer to the number of independent stress cycles. Third, the locked test period includes both a substantial upswing and the 2026 decline. Persistent defense reduced participation during favorable periods, while the later shock did not supply enough independent events to offset that cost.

Fourth, thresholds and weights were not retuned during the test period, by design. A more extensively tuned controller could almost certainly improve retrospective metrics, but such improvement would not establish mechanistic validity. Failure under a frozen specification is more informative than repeated tuning on the same history without a new external sample.

5.3. Alternative Explanations

Allostatic load may be merely a smoothed combination of volatility, drawdown, and duration. H3 rules out this explanation only if the composite state improves both scoring and calibration; the observed results do not meet that condition. Likewise, the fast-slow structure may simply be an ordinary multi-window filter. Because H2 failed, the current evidence cannot support a stronger claim.

Recovery memory may also be equivalent to a cooldown rule. Deleting it improved risk metrics, indicating that the current hysteresis implementation imposed a cost without providing protection. Dynamic quantile boundaries are formally valid probabilistic forecasts, but undercoverage requires caution when using the term "viability boundary". A boundary does not become protective merely because it changes dynamically; it must be calibrated and must produce identifiable consequences through better control decisions.

Parameter complexity does not explain why BARVC is favorable. The complexity-matched nonbiological model earned a higher return but assumed more risk, whereas the risk-matched constant-exposure model was simpler and performed better. The reversed model's lower mean exposure confounded the directional test, revealing a weakness in the negative-control design that must be corrected *ex ante* in future research.

5.4. Implications for Biomimetic Research Methods

The study's main methodological value lies in its falsification chain: each biological function corresponds to a module, a control action, an ablation, and a rejection condition. The results show why this chain matters. Without ablations and the constant-risk benchmark, the reduction in maximum drawdown from 36.08% to 21.77% could easily be presented as evidence of allostatic regulation. With appropriate controls, it primarily demonstrates the effect of lower risk exposure.

Future computational biomimetics should therefore meet at least four standards. First, distinguish functional translation from metaphorical naming. Second, compare the mechanism to the simplest intervention that can replicate the mean resource allocation. Third, prespecify mechanism direction and test it with a risk-exposure-matched reversed mechanism. Fourth, treat a negative module test as valid evidence about the translated mechanism rather than as an inconvenient tuning failure.

5.5. Financial and Control Implications

The experiment reaffirms a basic financial principle: a controller can manage risk exposure, but it cannot manage the external price. Forecast quality, the control mapping, transaction costs, and re-entry rules jointly determine capital viability. Lower predictive loss does not guarantee a better capital outcome; when mean exposure differs, lower drawdown does not prove adaptive capacity.

The P10-P90 results also caution against interpreting a 12-month fan chart as a deterministic target range. Under a well-calibrated model, P10 means that approximately 10% of outcomes fall below that value, conditional on the model and its inputs; P90 has the analogous upper-tail interpretation. The study's 20-day P10-P90 interval was undercovered, so its nominal 80% interpretation did not hold in the locked test period.

5.6. Strengths and Limitations

Strengths include a thirty-year audit trail, an explicit adjustment basis, event consolidation, temporal purging, locked testing, proper quantile loss, transaction costs, strong benchmarks, module ablations, reversed control, and full retention of negative findings. The study also explicitly separates retrospectively identified upward-move patterns from prospective predictive accuracy.

The limitations are equally material. The analysis covers one security, in which corporate actions, the construction-machinery cycle, policy, commodity demand, and market-wide shocks are intertwined. Daily dependence reduces the effective sample size, and the locked test period contains few recovery events. Forward-adjusted prices are suitable for return analysis but cannot exclude every source-data error. The load and memory equations represent only one of many possible translations. The model estimates three quantiles rather than a full predictive distribution. Additive calibration may be inadequate under rapid distribution shift. Transaction costs are simplified. Although constant 45% exposure was a prespecified round-number risk level, a fully prospective replication should determine the level from training and calibration exposures. The reversed controller was not exposure-matched. Market and industry controls, a cross-sectional security pool, delisted firms, and international validation are also absent.

5.7. Research Roadmap and Sample Expansion

The next stage should freeze the current controller and validate it on 300-500 A-share securities stratified by industry, market capitalization, liquidity, and listing cohort. Historical constituents and delisted securities are required to reduce survivorship bias. The primary sampling units for mechanism inference should be independent stress-recovery events, nested within securities, rather than trading days. A hierarchical model could estimate security-level heterogeneity while retaining the dependence structure among events.

The next validation stage should hold the current controller fixed and evaluate it across 300–500 A-share securities stratified by industry, market capitalization, liquidity, and listing cohort. The validation sample should include historical constituents and delisted securities to reduce survivorship bias. For mechanism inference, the primary sampling units should be distinct stress–recovery events nested within securities rather than individual trading days. A hierarchical model could estimate security-level heterogeneity while explicitly accounting for dependence among events within the same security.

6. Conclusions

This study translates physiological allostasis into a computational architecture for artificial risk regulation, assigning predictive inputs, fast and slow states, accumulated load, recovery memory, dynamic quantile boundaries, and bounded risk exposure to explicit deletion and rejection tests. Using 7,109 daily XCMG observations, the study reproduces 91 independent drawdown cycles, a historical maximum drawdown of 89.66%, and 124 retrospectively identified upward-move events.

During the 2022–2026 locked test period, BARVC reduced severe capital drawdown relative to buy-and-hold. Still, its P10–P90 interval was undercovered, and a constant 45% exposure model with a similar risk budget performed better. The results therefore supported none of the five prespecified hypotheses concerning anticipatory regulation, multi-timescale coordination, the incremental validity of allostatic load, recovery memory, and the superiority of the complete mechanism. This study demonstrates the feasible translation of biological functions into an auditable and falsifiable artificial controller; however, it does not demonstrate that this particular implementation improves investment performance.

The defensible conclusion is case-bounded: functional translation from physiological allostasis to a nonstationary artificial risk system is computationally feasible, but its empirical advantage is unconfirmed. Broader claims require external validation across securities and markets, with risk exposure matching.

Declarations

Author Contributions

Wei Meng independently completed the conceptualization, methodology, software, validation, formal analysis, data curation, visualization, and writing.

Funding

This study received no external funding.

Institutional Review Board Statement

Not applicable. The study uses market time-series data and involves no human participants, animals, clinical interventions, or identifiable personal information.

Informed Consent Statement

Not applicable.

Data Availability Statement

The analysis uses publicly available market data obtained through an open-source data interface. Redistribution of raw data is governed by the original provider’s license and terms. A reproducibility package may include the data dictionary, feature definitions, event rules, model specification, random seeds, derived states, test predictions, risk-exposure paths, and aggregate results.

Code Availability Statement

The analysis code fixes the temporal splits, purge rules, feature construction, model settings, controller logic, cost assumptions, and output tables. We will add a repository identifier after anonymous review is complete and the applicable license conditions have been verified.

Conflicts of Interest

No conflicts of interest are declared. This is a methodological study and does not constitute personalized investment advice or a promise of future performance.

Appendix A. Prespecified Variable Dictionary

Table 6 lists every variable entering the model and state recursions, its construction rule, and the latest time at which its information is available. The table is the principal index for checking temporal consistency and reproducing feature engineering; no variable may use information available after the close on day t .

Table 6. Prespecified variables, construction methods, and information timing

Domain	Variable	Construction method	Information timing
Momentum	5/20/60/120/252-day returns	$P_t/P_{t-k} - 1$	t (daily close)
Volatility	20/60-day realized volatility	Rolling standard deviation of daily returns $\times \sqrt{252}$	t (daily close)
Range risk	ATR/price	Wilder 14-day true range/closing price	t (daily close)
Capital stress	High-water-mark drawdown	Decline in current price from its historical high	t (daily close)
Duration	Time underwater	Consecutive trading days below the high-water mark	t (daily close)
Liquidity/activity	Abnormal volume; volume and turnover-rate ratios	Rolling transformations using historical data only	t (daily close)
Trend	Moving-average distance; Bollinger-band z score; trend strength	Rolling transformations using historical data only	t (daily close)
Tail stress	Breach frequency	Frequency of returns below the rolling P10 over the preceding 20 days	t (daily close)
Biomimetic states	F_t, S_t, L_t, H_t	Fast, slow, and load equations plus memory recursion	t (daily close)

Appendix B. Reproducibility Checklist

1. Fix the security and price-adjustment basis before modeling.
2. Sort by date and audit duplicates, missing values, negative volume/turnover values, and OHLC logic violations.
3. Fix the training, calibration, and test periods in chronological order.
4. Purge 20-day labels that cross temporal boundaries.
5. Estimate every transformation using training data only.
6. Freeze hyperparameters, state half-lives, weights, thresholds, and costs before the locked test.
7. Retain constant-exposure, reactive, complexity-matched, ablation, and reversed-control models.
8. Do not treat daily observations as independent; use a block bootstrap for paired daily differences.
9. Do not interpret retrospectively identified upward-move origins as prospective success probabilities.
10. Report all five unsupported hypotheses in full; do not replace them with post hoc hypotheses.

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